

Variance swaps as benchmarks for calibrating stochastic volatility models

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Abstract

This note derives the dynamics of forward variance curves implied by power-payoff options within an infinite-dimensional Itô framework. By expressing power-payoff prices through their implied variances and treating the forward variance curve as an L^2 -valued process, we obtain the full drift condition linking spot variance, volatility-of-volatility and maturity correlations. A key consequence is that the short-end forward variance equals the instantaneous spot variance. In the limit $p \rightarrow 0$, corresponding to log-contracts, the drift vanishes, yielding a particularly tractable forward-variance representation closely aligned with variance swap modelling practice.

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in I}, \mathbb{P})$ be a given filtered probability space satisfying the usual conditions (right-continuity and completeness) with a filtration $\{\mathcal{F}_t\}_{t \in I}$ defined over the index set $I = [0, T]$. To properly define stochastic integrals, we introduce the Hilbert space $\mathcal{H}^2(I \times \Omega, \mathcal{B}(I) \otimes \mathcal{F}, \mu \times \mathbb{P})$, where $\mathcal{B}(I)$ denotes the Borel- σ -algebra on I and μ denotes the Lebesgue measure, defined as the set of functions

$$\begin{aligned} f &: (I \times \Omega, \mathcal{B}(I) \times \mathcal{F}, \mu \times \mathbb{P}) &\rightarrow (\mathbb{R}^m, \mathcal{B}(\mathbb{R}^m)) \\ &(t, \omega) &\mapsto f(t, \omega) \end{aligned}$$

such that

1. f is progressively measurable w.r.t. $\{\mathcal{F}_t\}_{t \in I}$. That is, $f|_{[0, t] \times \Omega}$ is $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$ -measurable for each $t \in I$.
2. The condition

$$\int_I \mathbb{E} [|f(t, \omega)|^2] dt < \infty \quad (1)$$

holds. This implies that the trajectories $f(\cdot, \omega)$ are square-integrable functions $\mathbb{P}$ -a.s.

Identifying processes that are equal $\mu \times \mathbb{P}$ -a.s., the space $\mathcal{H}^2$ is a Hilbert space with respect to the norm

$$\|f\|_{\mathcal{H}^2} = \left[\int_I \mathbb{E} [|f(t, \omega)|^2] dt \right]^{1/2}. \quad (2)$$

At this point, we consider a financial market consisting of a riskless asset $B = \{B_t\}_{t \in I}$ and a risky asset $S = \{S_t\}_{t \in I}$, both adapted to the filtration $\{\mathcal{F}_t\}_{t \in I}$. The riskless asset price process is given by $B_t = B_0 e^{rt}$ for some constant interest rate $r \geq 0$. We take B as the numéraire and we fix an equivalent martingale measure¹, $\mathbb{P}^*$, under which the discounted price S_t/B_t is a martingale. Moreover, S_t is assumed to satisfy $\mathbb{E}^* [|S_t|^2] < \infty$. Let us assume that $\{\mathcal{F}_t\}_{t \in I}$ is the augmented filtration generated by an m -dimensional Wiener process $\{W_t : t \in I\}$ under $\mathbb{P}^*$. Consequently, applying Martingale Representation Theorem, we can write

$$\frac{S_t}{B_t} = \frac{S_0}{B_0} + \int_0^t \phi_s dW_s, \quad (3)$$

for some $\phi \in \mathcal{H}^2$. Thus, applying Itô's lemma

$$dS_t = rS_t dt + B_t \phi_t dW_t. \quad (4)$$

Following Bergomi's arguments on the characterisation of *usable* models for option pricing and hedging, the existence of payoff-independent break-even volatilities implies that the underlying process (assuming no jumps and the Markov property) must follow a diffusion process, as a result of the application of the Feynman-Kac theorem. In particular, under these conditions $\phi_t = \sigma_t \frac{S_t}{B_t}$. The process $\sigma_t \in \mathcal{H}^2$ is referred to as the instantaneous lognormal volatility (Bergomi 2016).

¹Although many stochastic volatility or interest-rate models (e.g. Heston or Hull-White) are incomplete, practical calibration to the full smile surface implicitly selects a particular equivalent martingale measure. See (Björk 2019) for a discussion.

We restrict ourselves to the study of power-payoff options. These are European options with maturity T and payoff $\Lambda : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow \mathbb{R}_{++}$ given by

$$\Lambda(S_T) = S_T^p, \quad (5)$$

where p is an arbitrary positive parameter.

If S_t is a Markov process under the risk-neutral probability measure $\mathbb{P}^*$, its price at time t must be a function only of $(t, S_t = s)$ (Shreve 2004), which we denote by $\text{Pow}_p(t, s)$. Using the properties of the Girsanov stochastic exponential, the price of a power-payoff option can be written as

$$\text{Pow}_p(t, s) = e^{-r(T-t)} \text{For}_S(t, T)^p \mathbb{E}^* \left[e^{\frac{p(p-1)}{2} \int_t^T \sigma_s^2 ds} \middle| \mathcal{F}_t \right], \quad (6)$$

where $\text{For}_S(t, T)$ denotes the forward price of S .

In full analogy with vanilla options, one may define an implied variance for power-payoff options, denoted by $\sigma_{\text{imp}, \text{Pow}_p}^2$ as the unique value that makes the Black-Scholes-style formula reproduce the market price:

$$\text{Pow}_p(t, s) = e^{-r(T-t)} \text{For}_S(t, T)^p e^{\frac{p(p-1)}{2} (T-t) \sigma_{\text{imp}, \text{Pow}_p}^2(t, T)}, \quad (7)$$

This definition is valid for all $p \neq 0, 1$ since the payoffs are strictly concave for $0 < p < 1$ and convex otherwise. Hence, standard no-arbitrage arguments guarantee that the price is finite and monotone in total implied variance. In particular, for any $T_1 < T_2$,

$$(T_1 - t) \sigma_{\text{imp}, \text{Pow}_p}^2(t, T_1) \leq (T_2 - t) \sigma_{\text{imp}, \text{Pow}_p}^2(t, T_2),$$

which ensures existence and uniqueness of $\sigma_{\text{imp}, \text{Pow}_p}$. Thus, implied variance is defined as the volatility parameter of the lognormal model that reproduces the observed price, exactly as in the vanilla case, and can be related to the instantaneous lognormal variance as

$$\sigma_{\text{imp}, \text{Pow}_p}^2 = \frac{2}{p(p-1)(T-t)} \log \mathbb{E}^* \left[e^{\frac{p(p-1)}{2} \int_t^T \sigma_s^2 ds} \middle| \mathcal{F}_t \right]. \quad (8)$$

Given the total implied variance of maturity T , we define the forward variance curve associated with power-payoff options by

$$\xi_t^{(p)}(T) = \frac{\partial}{\partial T} \left((T-t) \sigma_{\text{imp}, \text{Pow}_p}^2(t, T) \right). \quad (9)$$

Thus $T \mapsto \xi_t^{(p)}(T)$ represents the instantaneous variance rate implied by power-payoff options at maturity T , as viewed from time t .

Note that $\xi^{(p)}$ forms a random field $(t, T) \mapsto \xi_t^{(p)}(T)$. For a fixed maturity T , we assume that the process $t \mapsto \xi_t^{(p)}(T)$ admits a representation as an Itô process.

We now allow for a continuous dividend yield $q \geq 0$, so under the risk-neutral probability measure, $\mathbb{P}^*$, we postulate the dynamics

$$dS_t = (r - q)S_t dt + \sigma_t S_t dW_t^S \quad (10)$$

$$d\xi_t^{(p)}(T) = \mu_t^{(p)}(T) dt + \lambda_t^{(p)}(T) dW_t^T, \quad (11)$$

where the initial spot S_0 and the initial forward variance curve $\xi_0^{(p)}(T)$ are taken from market data. In practice, a continuous forward variance curve is not directly observed; instead, only a discrete set of market forward variances is available for a finite set of maturities. It is therefore common to interpolate between quoted maturities and to extrapolate the curve, in particular near the short end.

For each maturity T , equation (11) defines a one-dimensional Itô process driven by a Brownian motion $\{W^T\}_{t \in [0, T]}$. The family of Brownian motions $\{W^T\}_{T > 0}$ may be mutually correlated and also correlated with the Brownian motion W^S . For two maturities u, v , we denote

$$\rho_t(u, v) = \frac{d[W^u, W^v]_t}{dt}, \quad (12)$$

to the instantaneous correlation between W^u and W^v , and

$$\rho_t(S, u) = \frac{d[W^S, W^u]_t}{dt}, \quad (13)$$

to the instantaneous correlation between W^S and W^u .

Thus, for an uncountable set of maturities $T > t$, the collection $\{\xi_t^{(p)}(T)\}_{T > t}$ defines an infinite-dimensional Itô process. Rewriting the price process of the power-payoff option in terms of the forward variance curve, we have

$$\text{Pow}_p(t, s, \xi^{(p)}; T, r, q, p) = e^{-r(T-t)} \text{For}_S(t, T)^p e^{\frac{p(p-1)}{2} \int_t^T \xi_t^{(p)}(u) du}, \quad (14)$$

The martingale condition for $\text{Pow}_p(t, s, \xi^{(p)})/B_t$ under $\mathbb{P}^*$ can be written as

$$\mathbb{E}^* [d\text{Pow}_p(t, s, \xi^{(p)}) | \mathcal{F}_t] = r \text{Pow}_p(t, s, \xi^{(p)}) dt. \quad (15)$$

Note that Pow_p is a function of t and S_t , but a functional of the whole curve $u \mapsto \xi_t^{(p)}(u)$. As a consequence, the classical finite-dimensional Itô formula cannot be applied directly: the state variable includes the entire forward variance curve, which is an infinite-dimensional object, so Pow_p is not a $C^{1,2}$ function of an $\mathbb{R}^n$ -valued semimartingale, as required by the standard Itô formula. In this setting, one must resort to a functional Itô calculus based on Gâteaux derivatives with respect to the curve $\phi(\cdot) = \xi_t^{(p)}(\cdot)$ in a suitable function space.

For fixed t the truncated curve $\phi(u) := \xi_t^{(p)}(u) \mathbb{1}_{[t, T]}(u)$ is assumed to be an element of the Hilbert space $L^2([t, M])$. In this context, we write $\text{Pow}_p(t, s)[\phi]$ to emphasise that Pow_p is a function of (t, s) and a functional of the curve $\phi(\cdot)$. Therefore, the Itô formula takes the form

$$\begin{aligned} d\text{Pow}_p(t, s)[\phi] &= \frac{\partial \text{Pow}_p}{\partial t} dt + \frac{\partial \text{Pow}_p}{\partial s} dS_t + \frac{1}{2} \frac{\partial^2 \text{Pow}_p}{\partial s^2} d[S]_t \\ &\quad + \langle D_\phi \text{Pow}_p, d\phi_t \rangle + \frac{1}{2} \left\langle \left\langle D_\phi^2 \text{Pow}_p, d[\phi]_t \right\rangle \right\rangle + \left\langle D_\phi \left(\frac{\partial \text{Pow}_p}{\partial s} \right), d[\phi, S]_t \right\rangle. \end{aligned} \quad (16)$$

Functional derivatives act on test functions h through

$$\langle D_\phi \text{Pow}_p[\phi], h \rangle = \int_t^T \frac{\delta \text{Pow}_p}{\delta \phi(u)} h(u) du, \quad (17)$$

$$\left\langle \left\langle D_\phi^2 \text{Pow}_p[\phi], h \right\rangle \right\rangle = \int_t^T \int_t^T \frac{\delta^2 \text{Pow}_p}{\delta \phi(u) \delta \phi(v)} h(u, v) du dv, \quad (18)$$

$$\left\langle D_\phi \left(\frac{\partial \text{Pow}_p}{\partial s} \right), h \right\rangle = \int_t^T \frac{\delta}{\delta \phi(u)} \left(\frac{\partial \text{Pow}_p}{\partial s} \right) h(u) du. \quad (19)$$

Note that $D_\phi \text{Pow}_p[\phi]$ lies in the dual of $L^2([t, M])$, but since L^2 is a Hilbert space, we may canonically identify $L^2([t, M]) \cong L^2([t, M])^*$. Similarly, the second derivative $D_\phi^2 \text{Pow}_p[\phi]$ defines a bounded bilinear form on $L^2([t, M]) \times L^2([t, M])$, i.e., a bounded operator from L^2 to its dual space.

We now compute the contributions term by term.

(i) Time and spot derivatives. A straightforward computation gives

$$\begin{aligned}\frac{\partial \text{Pow}_p}{\partial t} &= \left(r - p(r - q) - \frac{p(p-1)}{2} \xi_t^{(p)}(t) \right) \text{Pow}_p, \\ \frac{\partial \text{Pow}_p}{\partial s} dS_t &= p(r - q) \text{Pow}_p dt + p \sigma_t \text{Pow}_p dW_t^S, \\ \frac{1}{2} \frac{\partial^2 \text{Pow}_p}{\partial s^2} d[S]_t &= \frac{p(p-1)}{2} \sigma_t^2 \text{Pow}_p dt.\end{aligned}$$

(ii) First functional derivative. By direct computation

$$D_\phi \text{Pow}_p[\phi] = \frac{p(p-1)}{2} \text{Pow}_p(t, s)[\phi],$$

and thus

$$\langle D_\phi \text{Pow}_p[\phi], d\phi_t \rangle = \frac{p(p-1)}{2} \text{Pow}_p(t, s)[\phi] \left[\left(\int_t^T \mu_t^{(p)}(u) du \right) dt + \left(\int_t^T \lambda_t^{(p)}(u) dW_t^u \right) \right].$$

(iii) Second functional derivative. Since

$$\frac{\delta^2 \text{Pow}_p}{\delta \phi(u) \delta \phi(v)} = \frac{p^2(p-1)^2}{4} \text{Pow}_p(t, s)[\phi],$$

and $d[\phi(u), \phi(v)]_t = \lambda_t^{(p)}(u) \lambda_t^{(p)}(v) \rho_t(u, v) dt$, we obtain

$$\frac{1}{2} \left\langle \left\langle D_\phi^2 \text{Pow}_p[\phi], d\phi_t \right\rangle \right\rangle = \frac{p^2(p-1)^2}{8} \text{Pow}_p(t, s)[\phi] \int_t^T \int_t^T \lambda_t^{(p)}(u) \lambda_t^{(p)}(v) \rho_t(u, v) du dv dt.$$

(iv) Cross-variation term. Since $\frac{\partial \text{Pow}_p}{\partial s} = \frac{p}{s} \text{Pow}_p$, we have

$$D_\phi \left(\frac{\partial \text{Pow}_p}{\partial s} \right) = \frac{p^2(p-1)}{2s} \text{Pow}_p(t, s)[\phi].$$

Using $d[\phi, S]_t = \lambda_t^{(p)}(u) \sigma_t S_t \rho_t(S, u) dt$,

$$\left\langle D_\phi \left(\frac{\partial \text{Pow}_p}{\partial s} \right), d[\phi, S]_t \right\rangle = \frac{p^2(p-1)}{2} \sigma_t \text{Pow}_p \int_t^T \lambda_t^{(p)}(u) \rho_t(S, u) du dt.$$

(v) **Final Itô expansion.** Summing all contributions and dividing by $\text{Pow}_p(t, s)[\phi]$ gives

$$\begin{aligned} \frac{d\text{Pow}_p(t, s)[\phi]}{\text{Pow}_p(t, s)[\phi]} &= \left[r + \frac{p(p-1)}{2}(\sigma_t^2 - \xi_t^{(p)}(t)) + \frac{p(p-1)}{2} \int_t^T \mu_t^{(p)}(u) du \right. \\ &\quad + \frac{p^2(p-1)^2}{8} \int_t^T \int_t^T \lambda_t^{(p)}(u) \lambda_t^{(p)}(v) \rho_t(u, v) du dv \\ &\quad \left. + \frac{p^2(p-1)}{2} \sigma_t \int_t^T \lambda_t^{(p)}(u) \rho_t(S, u) du \right] dt \\ &\quad + p\sigma_t dW_t^S + \frac{p(p-1)}{2} \int_t^T \lambda_t^{(p)}(u) dW_t^u. \end{aligned} \quad (20)$$

Since $\text{Pow}_p(t, s)/B_t$ must be a $\mathbb{P}^*$ -martingale, the drift of (20) must vanish, yielding

$$\begin{aligned} 0 &= \frac{p(p-1)}{2}(\sigma_t^2 - \xi_t^{(p)}(t)) + \frac{p(p-1)}{2} \int_t^T \mu_t^{(p)}(u) du \\ &\quad + \frac{p^2(p-1)^2}{8} \int_t^T \int_t^T \lambda_t^{(p)}(u) \lambda_t^{(p)}(v) \rho_t(u, v) du dv \\ &\quad + \frac{p^2(p-1)}{2} \sigma_t \int_t^T \lambda_t^{(p)}(u) \rho_t(S, u) du. \end{aligned} \quad (21)$$

Differentiating with respect to T gives the drift of the forward-variance curve:

$$\frac{p(p-1)}{2} \mu_t^{(p)}(T) + \frac{p^2(p-1)}{2} \sigma_t \lambda_t^{(p)}(T) \rho_t(S, T) + \frac{p^2(p-1)^2}{4} \int_t^T \lambda_t^{(p)}(T) \lambda_t^{(p)}(u) \rho_t(T, u) du = 0, \quad (22)$$

i.e.

$$\mu_t^{(p)}(T) = - \left[p\sigma_t \lambda_t^{(p)}(T) \rho_t(S, T) + \frac{p(p-1)}{2} \int_t^T \lambda_t^{(p)}(T) \lambda_t^{(p)}(u) \rho_t(T, u) du \right]. \quad (23)$$

Taking the limit $T \rightarrow t^+$ in (20), the only remaining terms are

$$\left[r + \frac{p(p-1)}{2} (\sigma_t^2 - \xi_t^{(p)}(t)) \right] dt,$$

and equating the drift to $r dt$ yields the short-end condition

$$\xi_t^{(p)}(t) = \sigma_t^2. \quad (24)$$

Thus, for every p , the instantaneous forward variance equals the instantaneous spot variance. The resulting dynamics are

$$\begin{aligned} dS_t &= (r - q)S_t dt + \sqrt{\xi_t^{(p)}(t)} S_t dW_t^S, \\ d\xi_t^{(p)}(T) &= - \left[p\sqrt{\xi_t^{(p)}(t)} \lambda_t^{(p)}(T) \rho_t(S, T) + \frac{p(p-1)}{2} \int_t^T \lambda_t^{(p)}(T) \lambda_t^{(p)}(u) \rho_t(T, u) du \right] dt + \lambda_t^{(p)}(T) dW_t^T, \end{aligned} \quad (25)$$

which is impractical to simulate in full generality, as it requires evolving the forward variance for every maturity separately and specifying an entire correlation structure across maturities.

Interestingly, the drift of (25) vanishes as $p \rightarrow 0$, which coincides with the definition of the log-contracts. Then, the dynamics of the log-contract forward variance curve is modeled by

$$\zeta_t^T = \xi_t^{(0)}(T) = \frac{d}{dT} [(T - t)\sigma_{\text{imp,LC}}^2(t, T)] .$$

Since log-contract prices are always finite under no-arbitrage, $\sigma_{\text{imp,LC}}$ is well-defined for all maturities. In the limit $p \rightarrow 0$, the drift of $\xi^{(p)}$ vanishes, leading to

$$dS_t = (r - q)S_t dt + \sqrt{\zeta_t^T} S_t dW_t^S, \quad (26)$$

$$d\zeta_t^T = \lambda_t(T) dW_t^T, \quad (27)$$

a remarkably convenient representation.

Although log-contracts provide an elegant theoretical route to characterising the forward variance curve, they require a continuum of vanilla options and are not traded in practice. Therefore, market participants instead use variance swaps. In practice, although the forward variance implied by log-contracts and the forward variance inferred from variance swaps may differ in the presence of jumps, they are usually treated as identical (Bergomi 2016). In that case, forward-variance models are then calibrated to the forward variance curve of variance swaps. Equivalently, as a consequence of the Eq. (8) when taking $p \rightarrow 0$,

$$\zeta_t^T = \mathbb{E}^* [\sigma_T^2 | \mathcal{F}_t], \quad (28)$$

stochastic volatility models can also be calibrated directly to the forward variance curve of variance swaps. These practical aspects will be addressed in future work.

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